Inclined Plane Problems

Answer key

Conventions used throughout

• **Forces:** F_g = gravity (F_g = mg, straight down), F_N = normal force (perpendicular to the ramp), F_T = tension, F_k = kinetic friction, F_s = static friction. F_{gx} and F_{gy} are the parts of gravity along and perpendicular to the ramp.

• **Axes (from the lesson):** +x points down the ramp (the direction of acceleration) and +y points perpendicular to the ramp, away from it. Then $a_y = 0$.

• **Signs in the working:** plain + and − numbers. **Directions in final answers:** [down the ramp], [up the ramp], [perpendicular into the ramp], [perpendicular out of the ramp].

• g = 9.81 m/s^2. Every step keeps full precision; only final answers are rounded.

STEPS

Choosing the x- and y-directions

2) To simplify the analysis, always choose the positive x direction to be in the direction of acceleration if possible. If you do this then the acceleration in the y-direction will be _____ (Due to Newton's Second Law)

The object moves along the ramp. It never sinks into the ramp or lifts off it, so it has no acceleration perpendicular to the ramp. If +x is along the acceleration, all of the acceleration is in x:

$$a_y = 0 \quad \Rightarrow \quad \Sigma F_y = m(0) = 0$$

FINAL ANSWER

Blank: zero (a_y = 0 m/s^2)

DRAWING FBDS

FBDs with x- and y-directions

Draw a FBD for the following situations, and label which direction for each component we want to be positive:

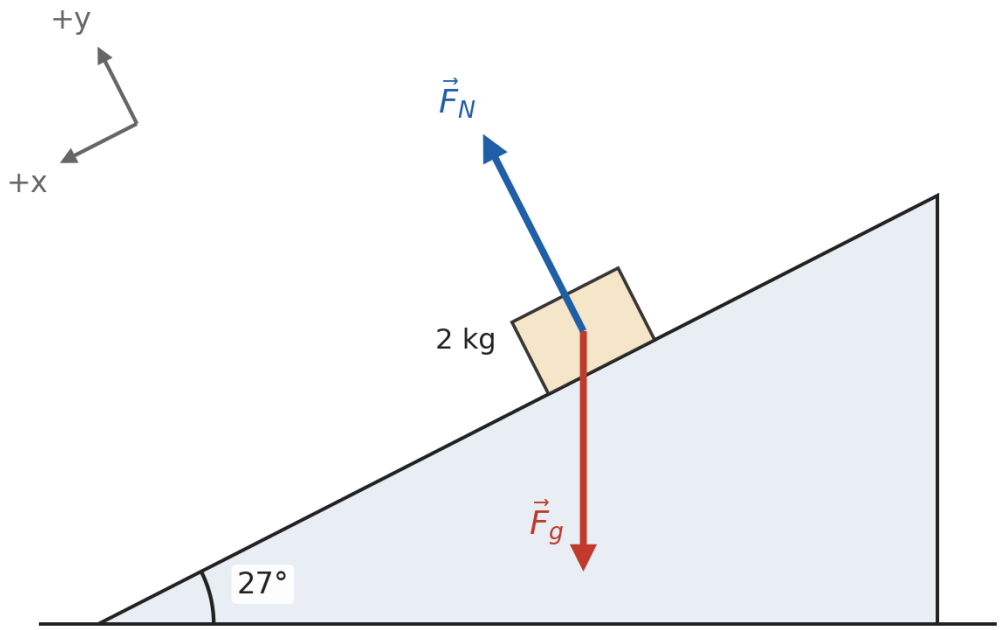
Example 1) A 2 kg box is put on the surface of an inclined plane at 27° above the horizontal (Assume the surface of the inclined plane to be frictionless).

Example 2) An object of 5kg rests on an inclined plane at 30° above the horizontal, with a rope holding it at rest on the plane.

Example 1 A 2 kg box is put on the surface of an inclined plane at 27° above the horizontal (Assume the surface of the inclined plane to be frictionless).

Two forces: gravity (straight down) and the normal force (perpendicular to the ramp). There is no friction. The box will slide down the ramp, so +x is down the ramp and +y is away from the ramp.

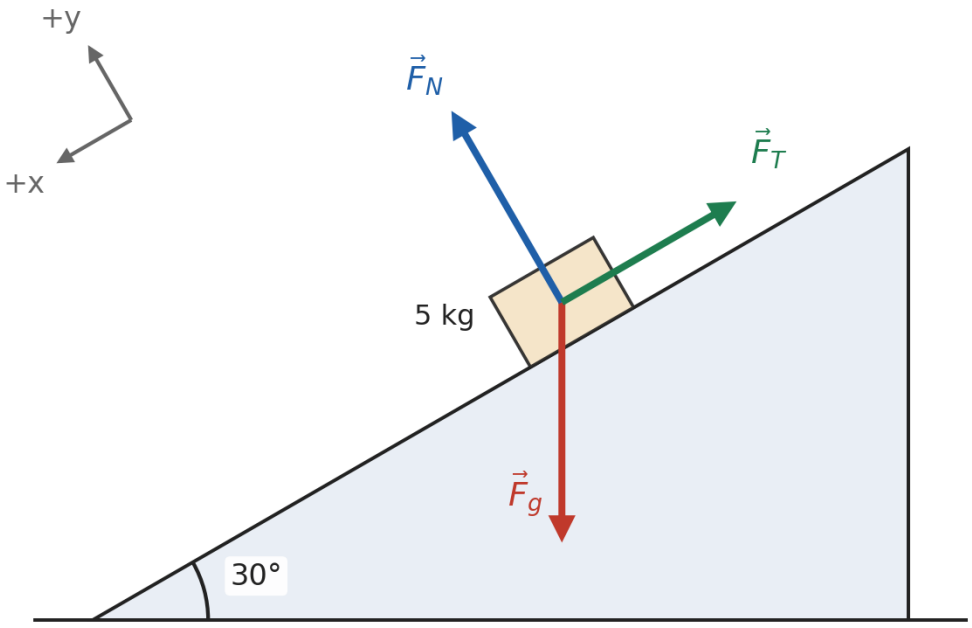
FBD: 2 kg box, frictionless 27° ramp



Example 2 An object of 5kg rests on an inclined plane at 30° above the horizontal, with a rope holding it at rest on the plane.

Three forces: gravity, the normal force, and tension along the rope (up the ramp). The object is at rest, so there is no acceleration direction; keep +x down the ramp (the way it would slide) and +y away from the ramp.

FBD: 5 kg object held by a rope on a 30° ramp



FINAL ANSWER

Example 1: F_g down, F_N perpendicular to the ramp; +x down the ramp, +y away from it

Example 2: add F_T up the ramp; same axes

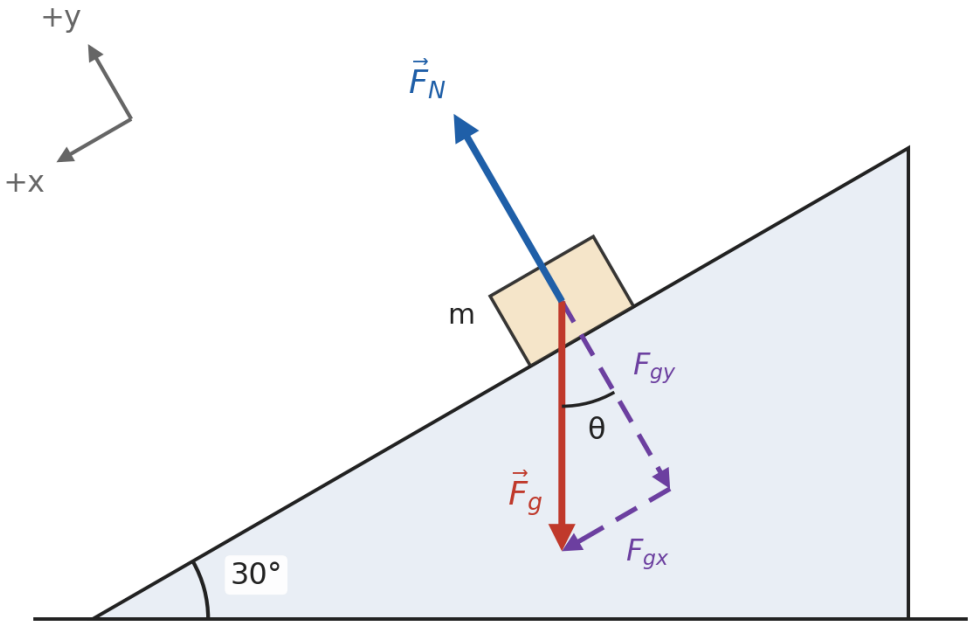
TYPICAL PROBLEM

Gravity on a ramp

1) What force in this diagram is responsible for the objecting sliding down the ramp? (Friction Negligible).

2) Using the diagram above, or re-drawing it on your paper, prove that the top angle in the force of gravity component triangle is the same as the angle of the ramp. In our last lesson, we assumed this to be true and took it for granted.

The typical inclined plane diagram (+x down the ramp, +y away from the ramp)



1) What force in this diagram is responsible for the objecting sliding down the ramp? (Friction Negligible).

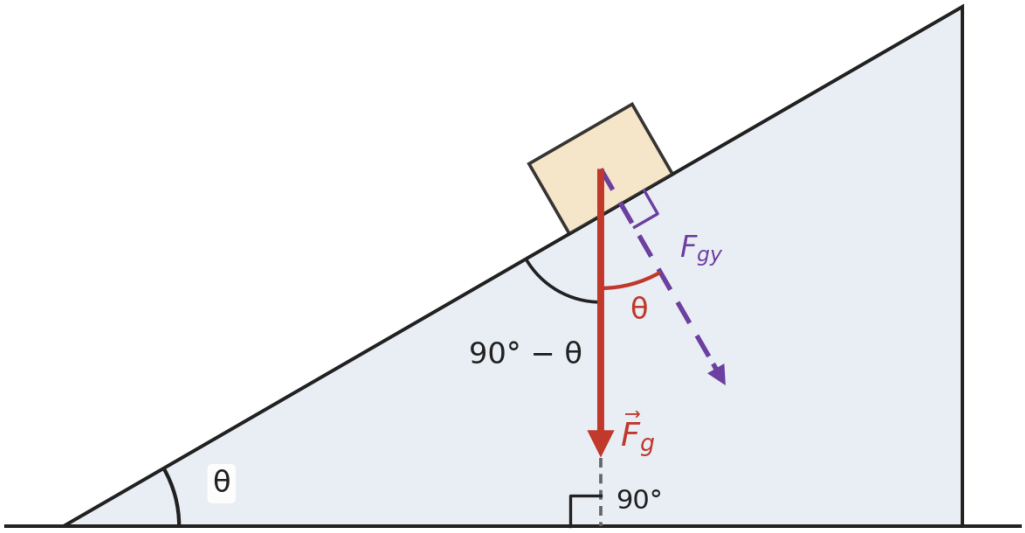
F_N is perpendicular to the ramp, and F_{gy} cancels it. The only force pointing along the ramp is F_{gx}, the part of gravity parallel to the ramp:

$$F_{gx} = mg\sin\theta$$

2) Prove that the top angle in the force of gravity component triangle is the same as the angle of the ramp.

Follow the angles around the diagram below. The dashed grey line continues F_g straight down to the ground.

Diagram: the angles



Step 1 – The big triangle (ramp, ground, vertical line)

The vertical line meets the ground at 90°, and the ramp meets the ground at θ. The angles in a triangle add to 180°, so the angle between the ramp and the vertical line is:

$$180° - 90° - \theta = 90° - \theta$$

Step 2 – The angle between F_g and F_{gy}

F_{gy} is perpendicular to the ramp, so it makes 90° with the ramp. F_g already makes 90° − θ with the ramp. The angle between F_g and F_{gy} is what is left:

$$90° - (90° - \theta) = \theta$$

So the top angle of the gravity triangle equals the ramp angle θ.

FINAL ANSWER

1) The component of gravity parallel to the ramp, $F_{gx} = mg \sin \theta$

2) The top angle is 90° − (90° − θ) = θ, the same as the ramp angle

EXAMPLE 1

4.2 kg box on a frictionless 30° ramp

Example 1) The box on the ramp above in our 'Typical Situation' has a mass of 4.2 kg and the ramp is angled at 30° above the horizontal.

a) Determine the component of gravity perpendicular to the ramp:

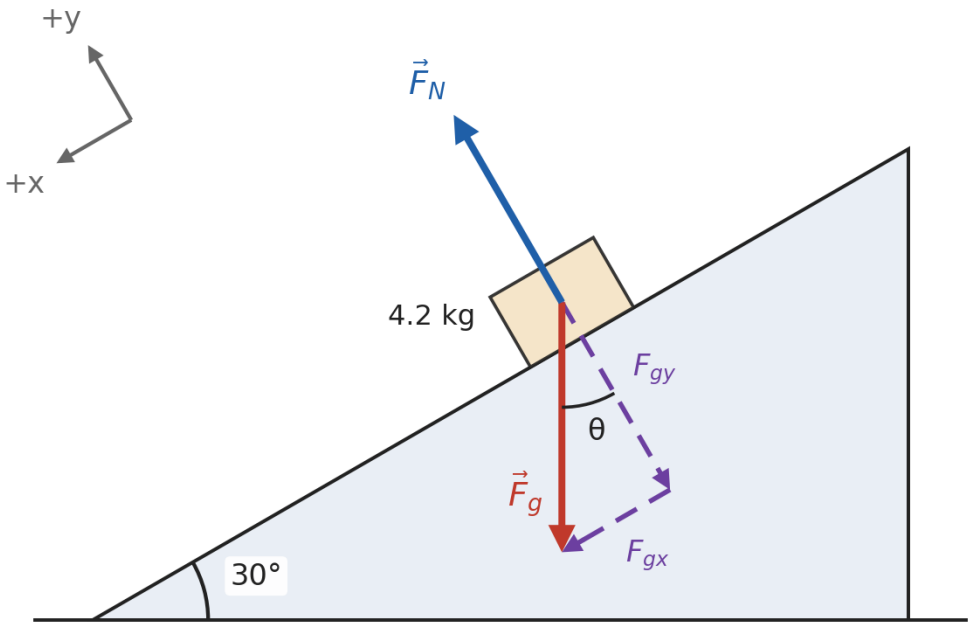
b) Why won't this component contribute to the acceleration?

c) Determine the component of gravity parallel to the ramp:

d) Solve for acceleration.

e) Determine the normal force.

FBD (+x down the ramp, +y away from the ramp)



GIVEN

$m = 4.2$ kg	$\theta = 30°$	$a_y = 0$	$F_{gy} = ?$
$F_{gx} = ?$	$a_x = ?$	$F_N = ?$	

(a) Determine the component of gravity perpendicular to the ramp:

F_{gy} is next to θ in the gravity triangle, so it uses cosine. It points into the ramp, which is −y:

Formula: $F_{gy} = -mg\cos\theta$

$F_{gy} = -(4.2)(9.81)\cos 30° = -35.68 \text{ N}$

(b) Why won't this component contribute to the acceleration?

F_{gy} is perpendicular to the ramp, but the box only moves along the ramp. The normal force pushes back with the same size, so the y forces cancel and $a_y = 0$.

(c) Determine the component of gravity parallel to the ramp:

F_{gx} is opposite θ, so it uses sine. It points down the ramp, which is +x:

Formula: $F_{gx} = mg\sin\theta$

$F_{gx} = (4.2)(9.81)\sin 30° = 20.60 \text{ N}$

(d) Solve for acceleration.

F_{gx} is the only force along the ramp, so it is the net force in x:

Formula: $\Sigma F_x = ma_x$

Put in $\Sigma F_x = F_{gx}$: $F_{gx} = ma_x$

Divide both sides by m: $a_x = \frac{F_{gx}}{m}$

$a_x = \frac{20.60 \text{ N}}{4.2 \text{ kg}} = 4.905 \text{ m/s}^2$

(e) Determine the normal force.

Formula: $\Sigma F_y = ma_y$

Put in the forces and $a_y = 0$: $F_N + F_{gy} = 0$

Put in $F_{gy} = -35.68$: $F_N - 35.68 = 0$

Add 35.68 to both sides: $F_N = 35.68$ N

FINAL ANSWER

(a) $\vec{F}_{gy}$ = 36 N [perpendicular into the ramp]

(b) The normal force cancels it, and the box doesn't move perpendicular to the ramp

(c) $\vec{F}_{gx}$ = 21 N [down the ramp]

(d) $\vec{a}$ = 4.9 m/s^2 [down the ramp]

(e) $\vec{F}_N$ = 36 N [perpendicular out of the ramp]

EXAMPLE 2

Adding kinetic friction

Example 2) Imagine the same problem as Example 1 above, but now we have friction involved.

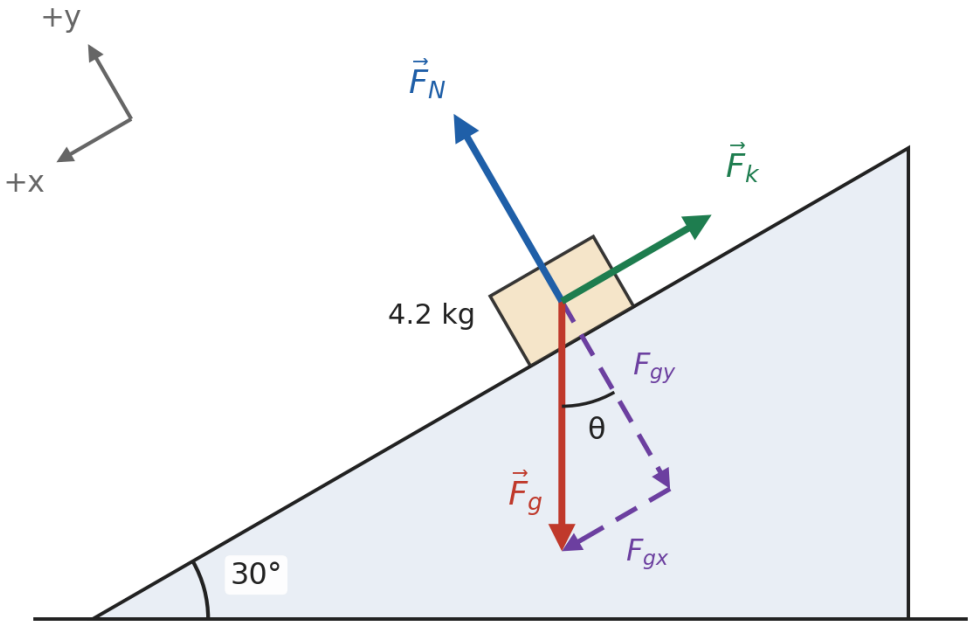
a) What would the free body diagram of the ramp look like now?

b) The constant of kinetic friction on the ramp uk = 0.27. (Recall that friction can be found using $F_k = \mu_k F_N$.) Determine the new acceleration of the box with friction included.

(a) What would the free body diagram of the ramp look like now?

Add kinetic friction to the box's FBD. The box slides down the ramp, so friction points up the ramp.

FBD with friction (+x down the ramp, +y away from the ramp)



(b) Determine the new acceleration of the box with friction included.

Step 1 – Find the friction force

Friction acts along the ramp, so the y forces are the same as Example 1 and F_N = 35.68 N. Friction points up the ramp, which is −x:

Formula: $F_k = \mu_k F_N$

$$F_k = (0.27)(35.68\ \text{N}) = 9.634\ \text{N} \quad \Rightarrow \quad F_{kx} = -9.634\ \text{N}$$

Step 2 – Solve for the acceleration

Formula: $\Sigma F_x = ma_x$

Put in the forces: $F_{gx} + F_{kx} = ma_x$

Divide both sides by m: $a_x = \dfrac{F_{gx} + F_{kx}}{m}$

$$a_x = \frac{20.60 + (-9.634)}{4.2} = \frac{10.97}{4.2} = 2.611\ \text{m/s}^2$$

Friction takes away almost half of the downhill push, so the acceleration drops from 4.9 to 2.6 m/s².

FINAL ANSWER

(a) F_g down, F_N perpendicular to the ramp, F_k up the ramp

(b) $\vec{a}$ = 2.6 m/s^2 [down the ramp]

INTRO TO FRICTION

Static friction on ramps

Let us re-do the FBD's for the first examples we did under "Drawing FBD with our X- and Y-Directions- Classroom Examples, but now we are considering friction.

Example 1) A 2 kg box is put on the surface of an inclined plane at 27° above the horizontal.

Example 2) An object of 5kg rests on an inclined plane at 30° above the horizontal, with a rope holding it at rest on the plane.

Discussion:

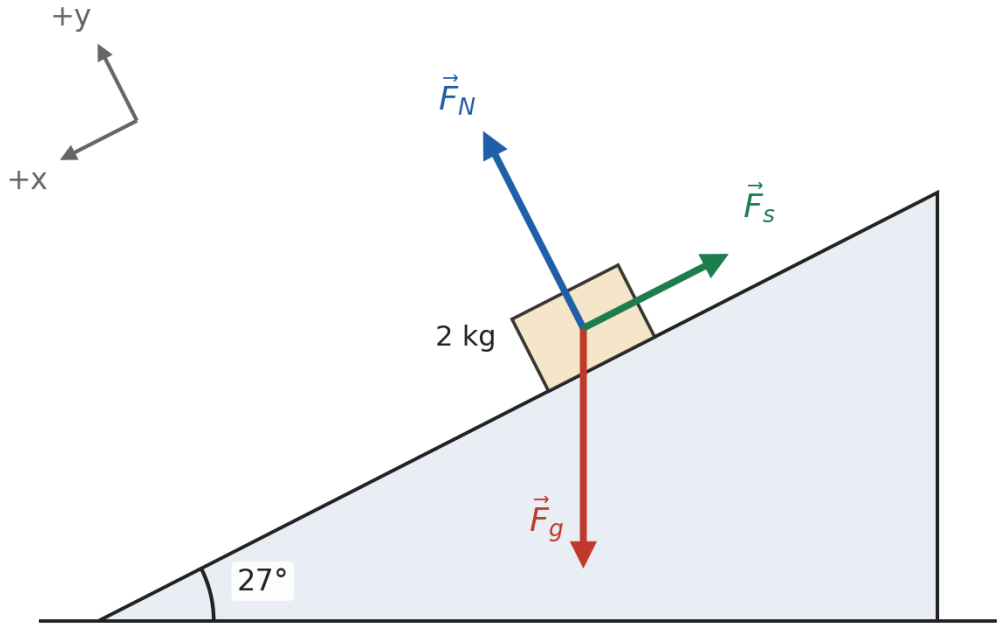
1) What component now would balance the force of gravity?

2) How does the force of static friction on the steeper ramp compare to the force of static friction of the shallower ramp? Explain.

Example 1 A 2 kg box is put on the surface of an inclined plane at 27° above the horizontal.

The box stays at rest, so the friction is static friction. Gravity tries to slide it down the ramp, so static friction points up the ramp.

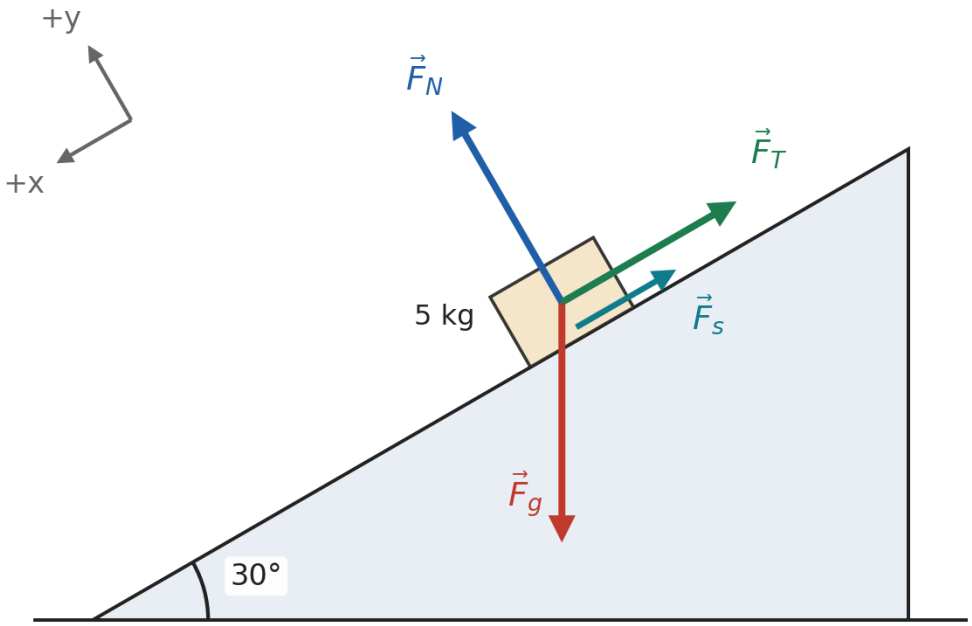
FBD: 2 kg box on a 27° ramp, with friction



Example 2 An object of 5kg rests on an inclined plane at 30° above the horizontal, with a rope holding it at rest on the plane.

The rope and static friction both point up the ramp and share the job of holding the object (friction only adds what the rope doesn't supply).

FBD: 5 kg object with a rope on a 30° ramp, with friction



Discussion 1) What component now would balance the force of gravity?

F_N balances F_{gy} (perpendicular). Along the ramp, static friction balances F_{gx}, so the box stays at rest. For the box with no rope (+x down the ramp):

Formula: $\Sigma F_x = ma_x$

Put in the forces and $a_x = 0$: $F_{gx} + F_{sx} = 0$

Subtract F_{gx} from both sides: $F_{sx} = -F_{gx} = -mg\sin\theta$

Discussion 2) How does the force of static friction on the steeper ramp compare to the force of static friction of the shallower ramp? Explain.

Static friction must equal $F_{gx} = mg \sin\theta$, and $\sin\theta$ is bigger for a steeper ramp. For the same 2 kg box:

$$27°: \quad F_s = (2)(9.81)\sin 27° = 8.91 \text{ N}$$

$$30°: \quad F_s = (2)(9.81)\sin 30° = 9.81 \text{ N}$$

The steeper ramp needs more static friction. The static friction force grows to match, but only up to its maximum, $F_{s,max} = \mu_s F_N$. If the ramp is steep enough that F_{gx} is bigger than that, the box starts to slide. (μ_s itself stays the same; it is the friction force that changes.)

FINAL ANSWER

1) Static friction, up the ramp, balances F_{gx} (the normal force balances F_{gy})

2) The steeper ramp needs more static friction, because $F_{gx} = mg \sin\theta$ is bigger